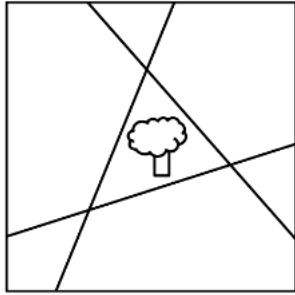


START for ALL PARTICIPANTS**1. THE PARK** (coefficient 1)

The park is divided into seven plots by three paths which run in a straight line from one side of the park to another. An old tree is already planted in the middle. For each of the three paths, there must be as many trees on one side as on the other.



How many young trees will you need to plant, at a minimum?

2. MILK CHURNS (coefficient 2)

Identical churns are filled with Swiss milk. Milk filling a certain volume always has the same weight. An empty churn and a half-filled churn of milk together weigh as much as a full churn of milk.

How many churns half filled with milk weigh, together, as much as two churns completely filled with milk?

3. PIN CODE (coefficient 3)

Giles wants Elisabeth to guess a four-digit code, which uses all the digits from 1 to 4. Elisabeth successively tries 1234, 2143 and 3124. According to the indications provided by Giles, at each of the three attempts, exactly two digits are in the same place as in the code.

What's the code?

4. EXAM ROOM (coefficient 4)

In the exam room holding the championship final, all the seats are occupied. There are the same number of participants in each row. Before settling down, Sandra counted there was one row behind her seat, and five rows in front of

her seat. From her place, she has two participants on her left and three on her right.

How many participants are there in the room?

5. ONE TO NINE (coefficient 5)

1								9
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Write all the integer numbers from 1 to 9 in the boxes, one number per box.

Each number, except the one on the far right, must divide into the sum of the numbers written to its right. The numbers 1 and 9 are already written. Just to the left of 9, we must therefore write 3, because 3 is the only remaining number that divides into 9.

END for CE PARTICIPANTS**6. QUEUEING** (coefficient 6)

On an island in Lake Geneva, nine people are waiting, one behind the other, for a boat ride. The queue is made up of two kinds of people, the liars who always lie and the sincere who always tell the truth. Each of the nine people says: "there are strictly more liars in front of me than sincere ones behind me".

How many liars are in the line?

7. PURSE (coefficient 7)

Scrooge only has twelve coins in his purse, and nothing else. If he randomly pulls out ten coins from his purse, he will always withdraw at least one 5 Swiss franc coin, two 2 Swiss franc coins and three 1 Swiss franc coins.

How many Swiss francs does Scrooge have in his wallet, in total?

8. MAKE THE CUTS (coefficient 8)

The same nine-digit number is printed on several strips of paper. It uses all the digits from 1 to 9. It begins with 36 on the far left. Colin cuts one strip of paper in two places in order to obtain three three-digit numbers whose sum is equal to 999. Corinne cuts another strip in three places to obtain from left to right 36, a number of three digits then two numbers of two digits. The sum of the four numbers is also equal to 999.

What is the nine-digit number?

END for CM PARTICIPANTS

Problems 9 to 18: beware! For a problem to be completely solved, you must give both the number of solutions, AND give the solution if there is only one, or give any two correct solutions if there are more than one. For all problems that may have more than one solution, there is space for two answers on the answer sheet (but there may still be just one solution).

9. MAKING SOME POINTS (coef. 9)

Several points have been marked on a line. At each step, between each pair of adjacent points, we mark a new point. After 9 such steps, 1537 points in total are marked.

How many points were marked before the first step?

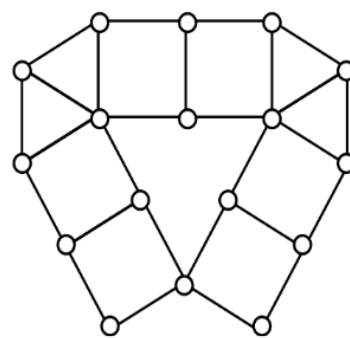
10. PRIORITY ORDERS (coefficient 10)

A, B and C are strictly positive integers. Lucas wants to calculate $(A + B) / C$. He types $A + B / C$ on his calculator and reads the result 25. He tries again as a check. He types $B + A / C$ and reads the result 35. Lucas then understands that his calculator gives priority to division over addition. In the first case, it divided B by C then added A. In the second case, it divided A by C then added B.

What is the correct result?

11. FUNERARY MASK (coefficient 11)

The drawing represents the funerary mask of Tutangent. The length of the sides of the six squares is equal to that

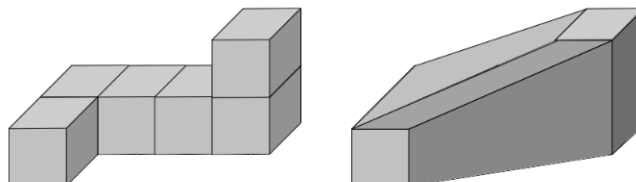


of the sides of the four equilateral triangles. The assembly is exact. Each circle represents a vertex, two or four vertices combined. In the centre, a fifth equilateral triangle is already formed.

How many equilateral triangles not already drawn can have three of the seventeen circles as vertices?

END for C1 PARTICIPANTS

12. THE LIBRARY IS PACKED (coef 12)



A campus library is composed of six cubes face-to-face, each with side 5 m. Hristo, a famous artist, has wrapped the building in plastic film.

What volume of air, in m^3 rounded to the nearest m^3 , is trapped between the building and the stretched film?

Note: if we connect the vertices of a polygon to a point, we obtain a pyramid whose volume is calculated by multiplying the area of the polygon by the distance between the point and the plane that contains the polygon, then dividing by 3.

13. DOUBLY TRUE (coefficient 13)



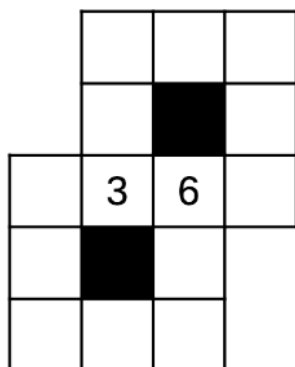
The scale above is balanced. Two different letters represent two different digits. All digits from 0 to 6 except one of them are used. J, E and L do not represent 0. The JM number is the exact result of dividing the EPF number by the L number. The weight of the "/" symbol is negligible. The balance is in equilibrium, which translates to $2J + M = E + 2P + 3F + 5L$ (the moments of the weights relative to the axis of rotation are equal).

Write each digit in its correct place.

14. ONE TO FOURTEEN (coefficient 14)

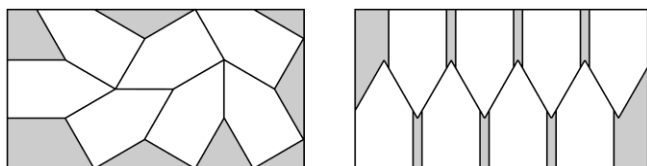
Write all the integers from 1 to 14 in the boxes, one per box.

The sum of the numbers on each of the four vertical alignments of three touching squares, on each of the two horizontal alignments of three squares and on the horizontal alignment of four squares must equal 25. The numbers 3 and 6 are already written.



END for C2 PARTICIPANTS

15. MARYNA'S CUCKOOS (coef. 15)



Maryna packs identical Swiss cuckoo clocks in rectangular parallelepiped cartons in two different ways, as seen from the side in the drawing.

In cross section, each cuckoo clock has the shape of a pentagon obtained by joining a square and an equilateral triangle with the same side length. The cuckoo clocks are packed and any contacts between them are flush: when two cuckoo clocks touch one another, they never do so at single, isolated points. Beware! There are empty spaces, shown in grey. Both cartons have the same vertical dimension as seen in the drawing. Although the carton on the left contains one fewer clock, the horizontal dimension seen of the carton on the left is greater than that of the carton on the right, by 12 mm exactly.

What is, in mm rounded to the nearest mm, the vertical dimension of the two rectangles as seen in the drawing?

Note: take $\sqrt{3}$ as 1.732.

Information: representing a chalet, the cuckoo clock is a pendulum clock that hangs on the wall. Its clockwork internals open to reveal a small wooden bird which opens its beak emitting a song at each passing hour, the famous "cuckoo".

16. A FRACTION OF A YEAR (coef. 16)

We express the fraction $\frac{36}{2022}$ as the sum of three different fractions.

The numerator of each fraction is equal to 1 and its denominator is a strictly positive integer. No other expression of this fraction as the sum of three different fractions uses a denominator strictly less than the smallest of the three denominators. In other words, ours is as small as possible.

What is the smaller of the other two denominators?

END for L1, GP PARTICIPANTS

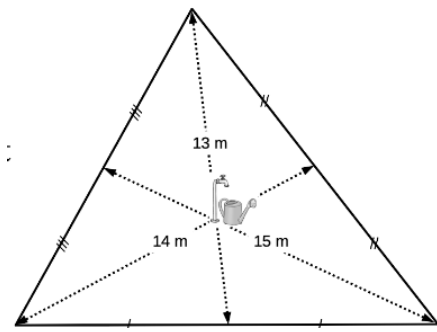
17. CUBES (coefficient 17)

A carpenter wants to cut a large cube of edge length 6 into small cubes of integer edge length. Between two cuts, he can select pieces and organize them so that he can saw them all together. A cut is made within a plane and without stopping mid-way through. For example, in order to obtain twenty small cubes, five cuts are necessary and sufficient: the carpenter then obtains a cube of edge length 4 and nineteen cubes of edge length 2.

In order to obtain sixty-four small cubes, how many cuts are necessary and sufficient?

18. PHIL McCAN's GARDEN (coef. 18)

Phil McCan's vegetable patch has the shape of a triangle divided into six plots by three paths, which each connect a vertex to the middle of the opposite side. They converge at a point where he has installed a watering tap.



The paths are 13, 14 and 15 m long.

What is the area of the patch in m^2 rounded to the nearest m^2 ?

END for L2, HC PARTICIPANTS